

A Gravitational Scaling Law for Atrial Re-Entrant Vulnerability: A Dimensionless Predictor of the Critical Gravity for Spaceflight Atrial Fibrillation

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Abstract

Background: Long-duration spaceflight remodels the atrium: a headward fluid shift dilates the chamber, and atrial electrophysiology changes in ways that raise markers of atrial fibrillation (AF) risk, although sustained AF has not been observed in astronauts. As permanent habitation of the Moon (0.16 g) and Mars (0.38 g) is planned, it is unknown whether a gravitational threshold exists below which the atrium acquires an arrhythmogenic substrate, and where the Moon and Mars lie relative to it. Cardiac physiology has scaling laws with body mass but none with gravitational acceleration.

Methods: We propose a dimensionless Cardiac Gravitational Number, $\mathcal{N}_g = \text{Lo-dilation}(g)/\text{WL}(g)$, where $\text{WL} = \text{CV} \cdot \text{APD}_{90}$ is the electrical wavelength, and re-entry can be hosted when $\mathcal{N}_g \geq 1$, defining a critical gravity g^* with $\mathcal{N}_g(g^*) = 1$. A validated human-atrial ionic model (Courtemanche–Ramirez–Nattel) was driven by a continuous, literature-anchored map from gravitational level to atrial remodelling, and $\mathcal{N}_g$ was computed across 0–1 g. The prediction was independently corroborated in a two-dimensional reaction–diffusion sheet by measuring rotor (phase-singularity) burden, and sensitivity to atrial size and map shape was quantified.

Results: $\mathcal{N}_g$ rose monotonically as gravity fell (0.47 at Earth to 1.29 in free fall), crossing unity at $g^* \approx 0.22$ — between the Moon ($\mathcal{N}_g \approx 1.07$, vulnerable) and Mars ($\mathcal{N}_g \approx 0.82$, protected). The full sheet reproduced the trend, with rotor burden zero at Earth and rising as gravity fell (gradual, not sharp, transition). g^* was robust to the assumed map shape (0.13–0.34) but strongly dependent on atrial size (no vulnerability at $\text{Lo} = 6$ cm; Mars vulnerable at $\text{Lo} = 12$ cm).

Conclusions: A simple dimensionless argument predicts a critical gravity for atrial re-entrant vulnerability between the Moon and Mars, corroborated by an independent model. Its strong dependence on atrial size yields a falsifiable, pre-flight-measurable hypothesis: individual atrial dimensions may set off-world arrhythmia risk.

Keywords: atrial fibrillation, microgravity, space medicine, cardiac electrophysiology, re-entry, mechano-electric feedback, in silico model, scaling law

Introduction

Cardiovascular adaptation is a recognised constraint on long-duration human spaceflight. In microgravity a headward fluid shift raises atrial filling and dilates the chamber; deconditioning and altered loading change atrial structure; and autonomic shifts alter atrial refractoriness. On Earth, each of these is independently implicated in the substrate for atrial fibrillation (AF). In astronauts after six-month missions, Khine et al. reported transient left-atrial enlargement ($+12 \pm 18$ mL) and electrophysiological changes consistent with increased AF risk, although no sustained AF was captured; AF prevalence in active astronauts ($\sim 5\%$) resembles the general population but occurs at a younger age [1].

As humanity plans sustained presence on the Moon (0.16 g) and Mars (0.38 g), a question follows that current evidence does not answer: is there a gravitational threshold below which the atrium acquires an arrhythmogenic substrate, and where do the Moon and Mars fall relative to it? Existing models treat spaceflight remodelling as a binary contrast (1 g versus “microgravity”) and are descriptive. Physiology offers powerful allometric scaling laws with body mass, but none with gravitational acceleration.

We treat gravity as a continuous variable and compress the question into a single dimensionless number, in the spirit of the governing numbers of fluid mechanics. We then test whether an independent, spatially resolved reaction–diffusion model agrees, and quantify how far the prediction can be trusted given its assumptions. This is a hypothesis-generating study: it proposes and tests a scaling argument *in silico* and does not claim experimental proof.

Materials And Methods

The mechanistic chain. Each link rests on established physiology: (i) on Earth a vertical hydrostatic column $\Delta P = \rho gh$ exists and vanishes as $g \rightarrow 0$, driving a cephalad fluid shift; (ii) increased preload raises atrial volume and, by the law of Laplace ($\sigma = P \cdot r / 2t$), wall stress and myocyte stretch — clinically, left-atrial enlargement in spaceflight [1]; (iii) atrial stretch shortens the refractory period and/or slows conduction velocity (CV) and increases their dispersion via mechano-electric feedback [2]; (iv) the electrical wavelength $WL = CV \cdot APD_{90}$ shortens until it no longer fits the available path length, and wavefronts fragment into re-entrant rotors.

The Cardiac Gravitational Number. We define $\mathcal{N}_g = L(g)/WL(g) = L_0 \cdot \text{dilation}(g)/(CV \cdot APD_{90}(g)/1000)$, with L_0 the ground atrial path length (human left-atrial circumference $\approx 8\text{--}12$ cm; nominal 8 cm) and $CV \approx 58$ cm/s. Re-entry can be hosted when $\mathcal{N}_g \geq 1$; the critical gravity g^* solves $\mathcal{N}_g(g^*) = 1$.

Continuous gravity $\rightarrow$ remodelling map. A minimal linear map sends g to atrial remodelling through the fluid-shift drive $(1 - g)$: electrical-remodelling severity, dilation, and fibrosis density each scale linearly with $(1 - g)$, calibrated so $g = 1$ recovers a validated ground baseline and $g = 0$ an established microgravity operating point.

Ionic model. APD_{90} was measured with the Courtemanche–Ramirez–Nattel (CRN) human-atrial action-potential model [3] at 1 Hz to steady state, with AF-type remodelling (reduced I_{CaL} , I_{to} , I_{Kur} ; increased I_{K1}) scaling with severity. All severities were paced in one vectorised run; the $g = 0$ column reproduced the scalar baseline $APD_{90} = 293.8$ ms exactly.

Full-sheet corroboration. A two-dimensional monodomain reaction–diffusion sheet (CRN ionics; fibre-anisotropic conduction calibrated to ≈ 58 cm/s; spatially correlated fibrosis) was

built at each gravity level with all tissue knobs driven by the same map. A broken wavefront was induced and rotor burden quantified as phase-singularity density via the topological-charge method on a time-delayed-voltage phase field.

Sensitivity. Without re-simulating the ionic model, g^* was recomputed for $L_0 \in \{6, 8, 10, 12\}$ cm and fluid-shift drive exponents $p \in \{0.5, 1, 2\}$ in $(1 - g)^p$. All code and parameters are openly available (see Additional Information).

Results

$\mathcal{N}_g$ rose monotonically as gravity fell (Table 1), crossing unity at $g^* \approx 0.22$. The Moon (0.16 g) lies just inside the vulnerable region ($\mathcal{N}_g \approx 1.07$); Mars (0.38 g) lies outside it ($\mathcal{N}_g \approx 0.82$); interplanetary transit (0 g) is most vulnerable ($\mathcal{N}_g = 1.29$); Earth is safe ($\mathcal{N}_g = 0.47$).

Table 1: Cardiac Gravitational Number across gravitational level ($L_0 = 8$ cm, $CV = 58$ cm/s).

g (Earth-g)	Body	APD ₉₀ (ms)	WL (cm)	$\mathcal{N}_g$
0.00	Interplanetary transit	139.3	8.08	1.29
0.16	Moon	≈ 163	≈ 9.4	≈ 1.07
0.38	Mars	≈ 202	≈ 11.7	≈ 0.82
0.50	—	219.8	12.75	0.72
1.00	Earth	294.1	17.06	0.47

The full reaction–diffusion sheet (6-seed ensemble per gravity level, bootstrap 95% CI) reproduced the direction of the law (Table 1b): rotor burden was comparably high across transit, the Moon and Mars ($\approx 9\text{--}12 \times 10^{-4}$), fell sharply between 0.5 and 0.8 g, and was exactly zero at Earth (CI 0–0). The models agree at the extremes and in direction but differ in where the transition completes: the analytic threshold is crisp at $g^* \approx 0.22$, while the spatially resolved model is more pessimistic, sustaining rotors through the Mars level and approaching zero only near 0.5–0.6 g. The single-cell argument is thus an optimistic bound on the vulnerable gravity range.

Table 1b: Full-sheet rotor burden (6-seed ensemble, mean and bootstrap 95% CI).

g (Earth-g)	Body	PS density $\times 10^{-4}$ (mean, 95% CI)
0.00	Interplanetary transit	9.05 (8.14–9.81)
0.16	Moon	11.51 (8.87–14.24)
0.38	Mars	10.85 (9.14–12.43)
0.50	—	4.76 (3.30–5.86)
0.80	—	0.21 (0.00–0.62)
1.00	Earth	0.00 (0.00–0.00)

g^* was robust to the assumed fluid-shift map shape ($g^* \in [0.13, 0.34]$ for $p \in \{0.5, 1, 2\}$) but depended strongly on atrial path length: no crossing (no vulnerability at any gravity) for $L_0 = 6$

cm, $g^* = 0.22$ at 8 cm, 0.41 at 10 cm, and 0.58 at 12 cm — at which point Mars itself becomes vulnerable (Table 2).

Table 2: Sensitivity of the critical gravity g^* .

Assumption	Value	g^*
$L_o = 6$ cm	Small atrium	None (always safe)
$L_o = 8$ cm	Nominal	0.22
$L_o = 10$ cm	Enlarged	0.41
$L_o = 12$ cm	Dilated	0.58 (Mars vulnerable)
Drive exponent $p = 0.5$	Concave	0.34
$p = 1.0$	Linear	0.22
$p = 2.0$	Convex	0.13

Discussion

A one-line dimensionless argument predicts a critical gravity for atrial re-entrant vulnerability between the Moon and Mars, and an independent reaction–diffusion model corroborates the trend. The prediction is consistent with the clinical picture — increased vulnerability without sustained AF [1] — because $\mathcal{N}_g$ indexes the substrate’s proximity to re-entry, not the occurrence of fibrillation.

The most decision-relevant result is the strong dependence on atrial size: because g^* moves from “no one is ever vulnerable” ($L_o = 6$ cm) to “even Mars is vulnerable” ($L_o = 12$ cm), the framework predicts that individual atrial dimensions could determine off-world arrhythmia risk — a personalised, pre-flight-measurable hypothesis, since atrial size is routinely imaged.

This study has clear limitations. The gravity → stretch link is the weakest: the map uses acute-stretch mechano-electric sensitivities as a stand-in for chronic adaptation, and the linear form is a hypothesis, not data. The corroboration uses a single ionic model in two dimensions, without autonomic dynamics or three-dimensional geometry, and is a direction-of-effect check rather than a calibrated incidence estimate. $\mathcal{N}_g$ is proposed, not proven; the analytic threshold is more optimistic about Mars than the full model. No human partial-gravity electrophysiology exists to fix the absolute value of g , so the headline $g \approx 0.22$ is conditional on $L_o \approx 8$ cm and the stated calibration. The framework is nonetheless falsifiable: partial-gravity measurements of atrial APD₉₀, CV and effective path length that yield an $\mathcal{N}_g$ not crossing unity within $[0, 1]$ g, or crossing it far from the predicted band, would refute the hypothesis.

Conclusions

A dimensionless Cardiac Gravitational Number predicts a critical gravity for atrial re-entrant vulnerability that falls between the Moon and Mars and is corroborated by an independent reaction–diffusion model. The prediction depends primarily on atrial size, giving a concrete, falsifiable, pre-flight-measurable hypothesis: that individual atrial dimensions may set off-world arrhythmia risk. The decisive test is partial-gravity atrial electrophysiology, ideally stratified by atrial size.

Additional Information

Author Contributions: LPR conceived the hypothesis, designed and implemented the model and analyses, produced the figures, and wrote the manuscript.

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Data Availability: All code, parameters, and scripts that reproduce every figure and number are openly available at <https://github.com/laurapiro17/atrial-fib-in-microgravity>.

References

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